SeeWandel Climate

Interreg
Alpenrhein-Bodensee-Hochrhein

Kofinanziert von der Europäischen Union

Regime Shifts in Plankton Dynamics: What Drives Them and Can We Predict Them?

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www.seewandel.org

Ecomeets @ Eawag, Dübendorf
28 October 2026

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The Pomati lab



Pomati lab
Plankton Ecology, Eawag

SeeWandel-Climate
Lake Constance

Take home message

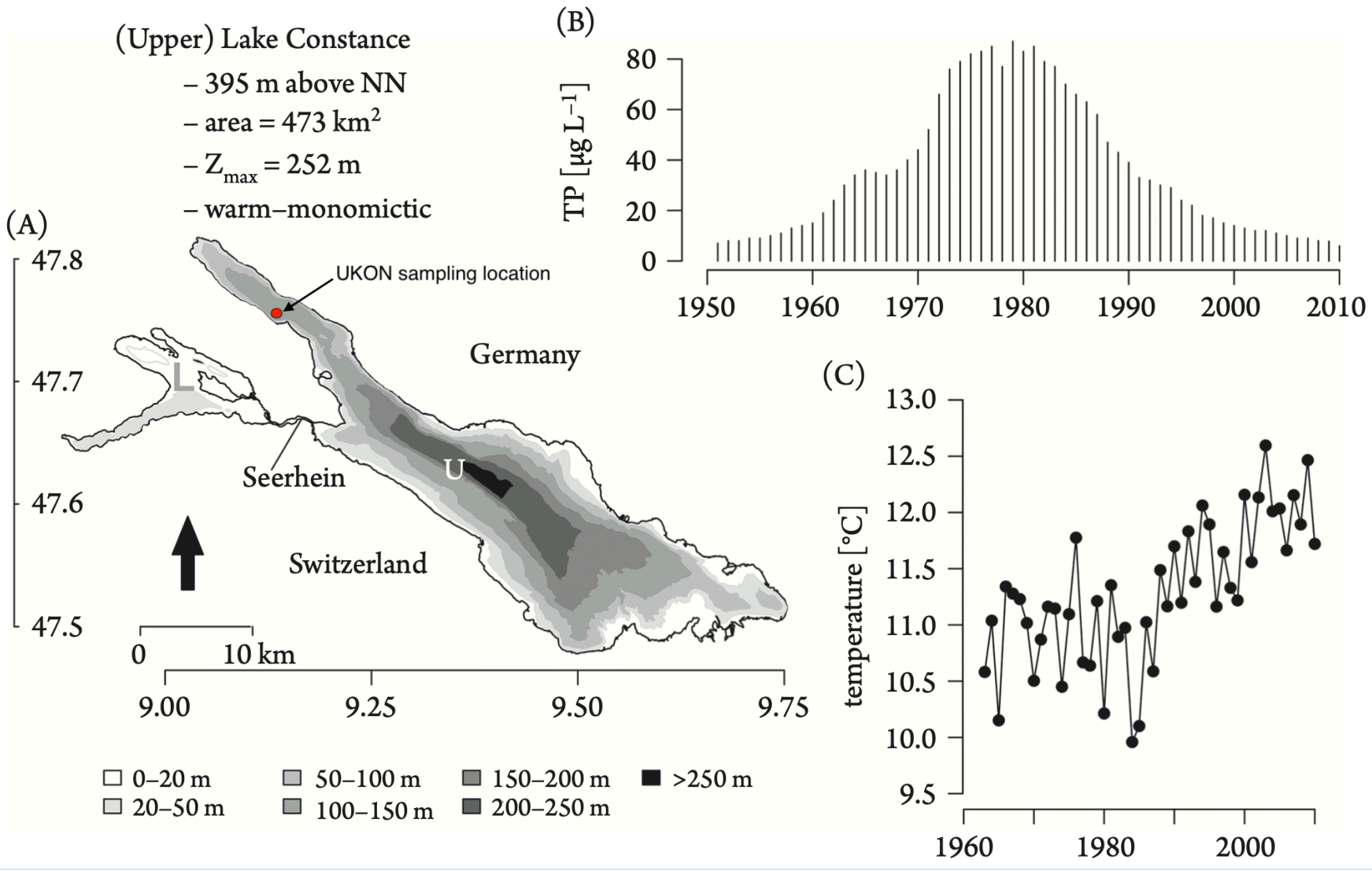
Plankton regime shifts can be detected and partly explained by biotic drivers. Whether they propagate and can be predicted is still open.



Outline

1. What we know: background
2. Research questions
3. Methods: our pipeline
4. Results: detection of regime shifts
5. Explain regime shifts
6. Propagation of regime shifts
7. Conclusions

What we know: background

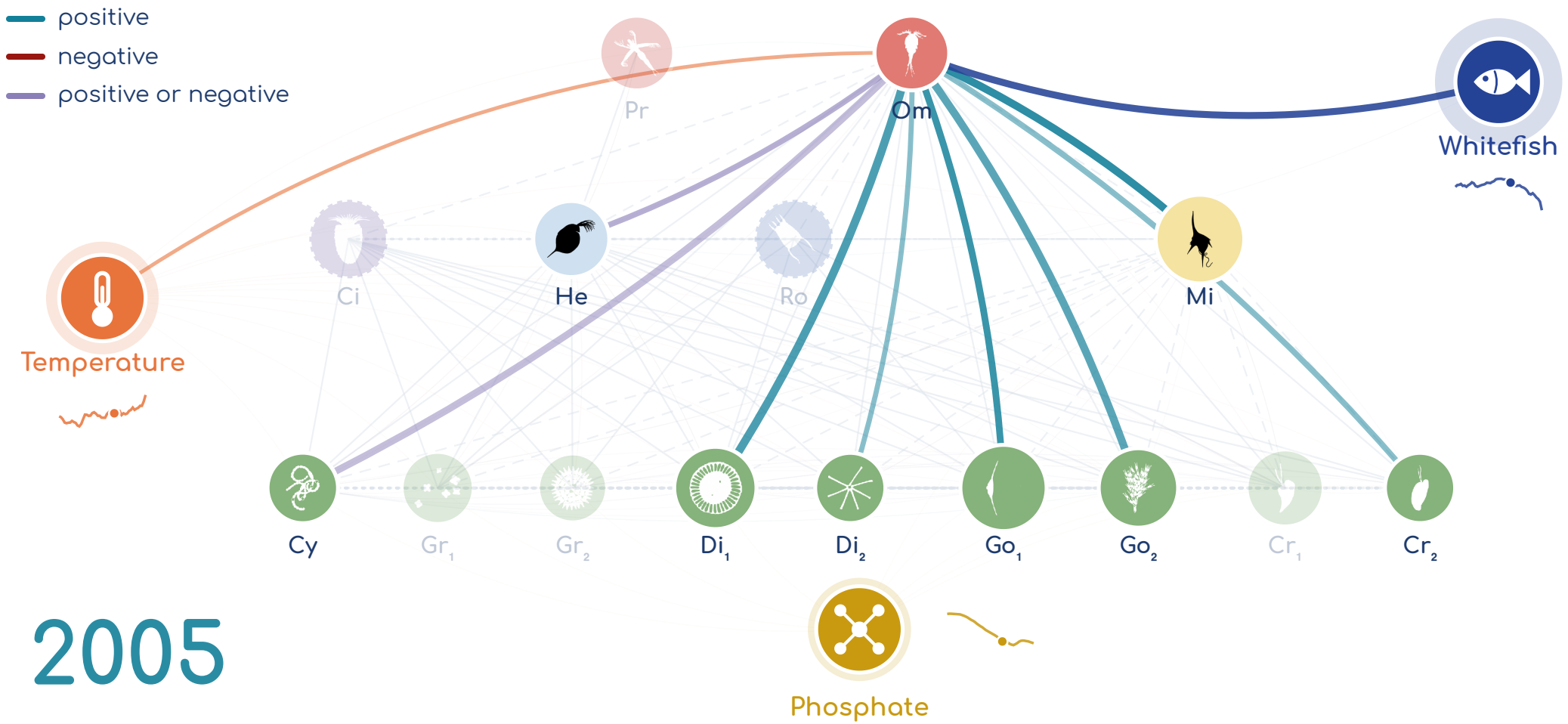


Lake Constance
473 km² · 252 m deep

Phosphate ↑ until 1980, then ↓

Temperature ↑ about 1 °C

What we know: the plankton community



2005

Spotlight: what we expect to drive the omnivores (Om)

Network modified from Merz et al. 2023, Nat. Clim. Change.

Research questions



1 · Detect

Abrupt shifts?



2 · Explain

What drives them?



3 · Propagate

Does a shift spread?



4 · Predict

Can we forecast them? (in progress)

Methods: our pipeline

1

Data

Monthly data 1975–2022

2

Detect

BEAST + two criteria

3

Explain

GP-EDM + hypotheses

4

Propagate

Shift-removal test

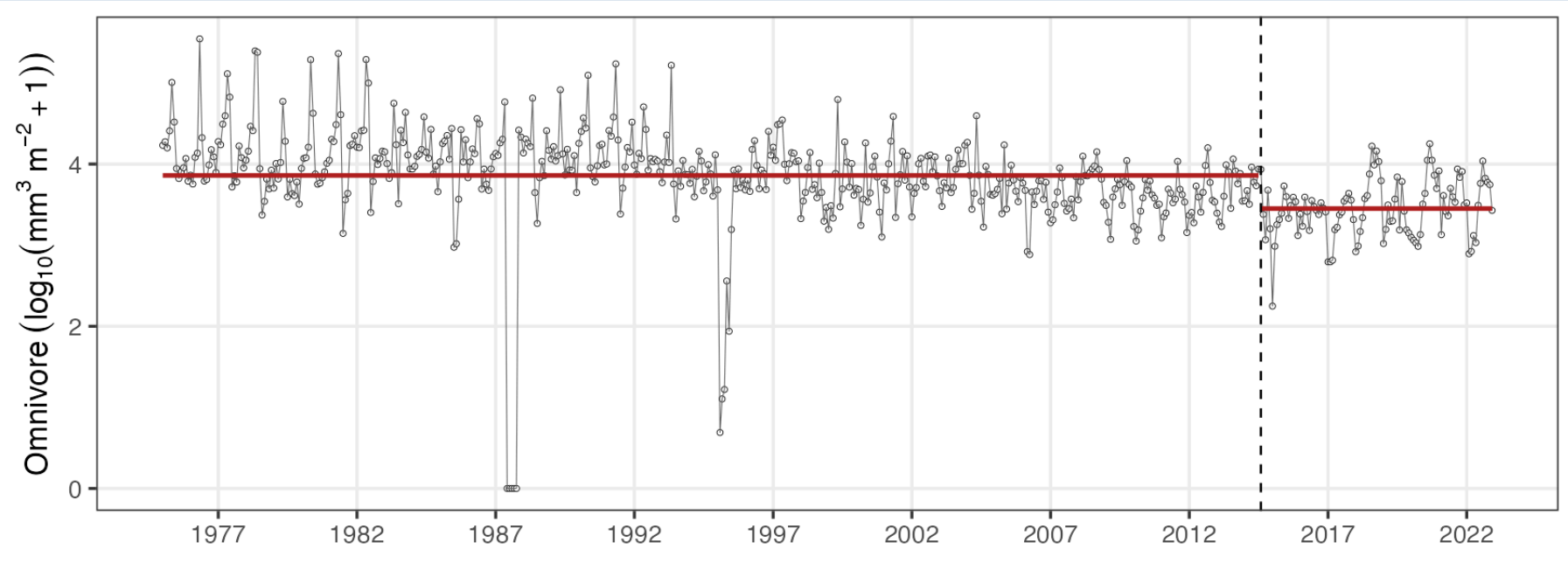
5

Predict

Forward validation

WORK IN PROGRESS

Results: detection of regime shifts



Omnivores, monthly biovolume. Dashed: shift; red: period means.

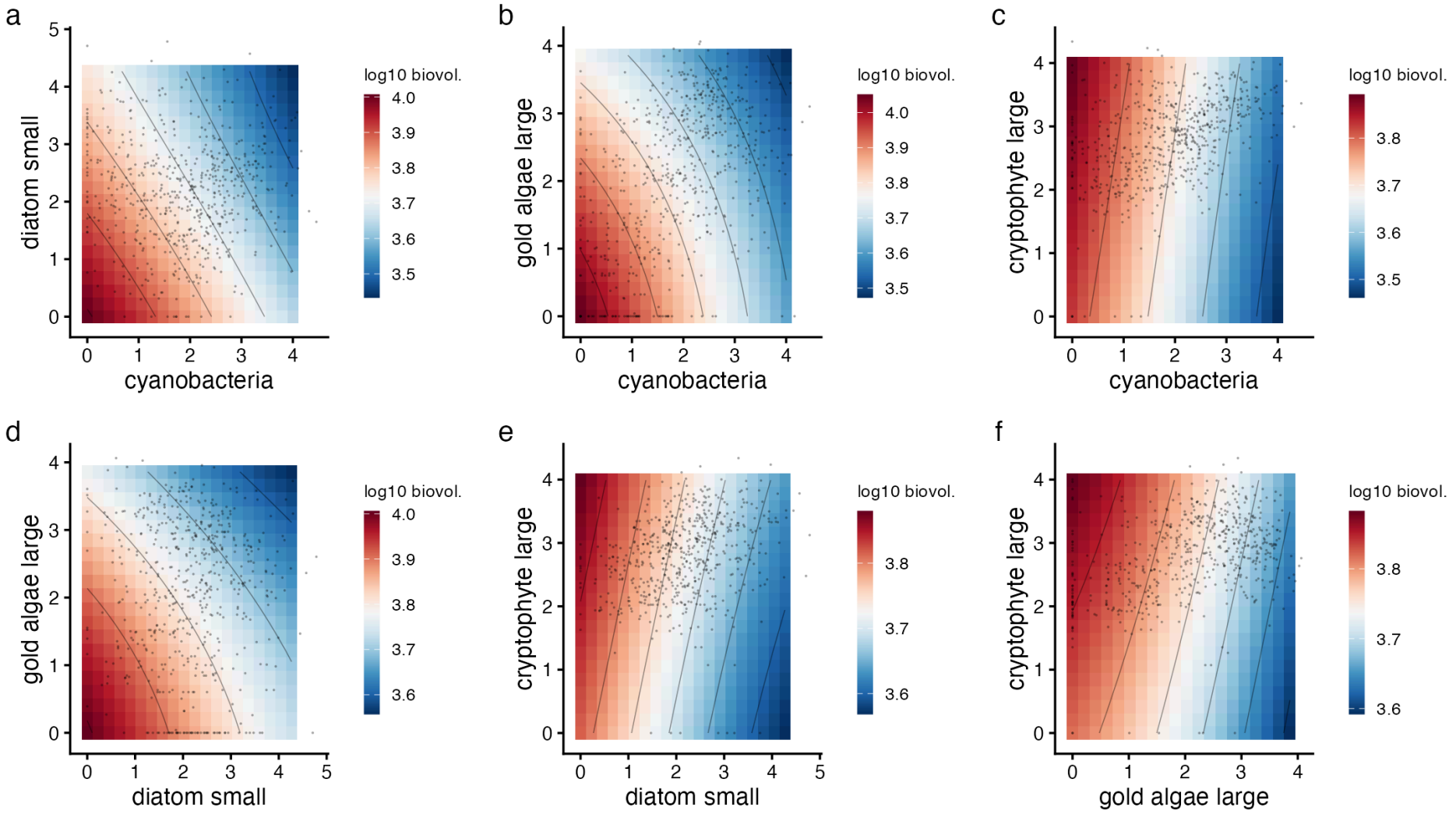


Shift: Aug 2014

abruptness 1.19

The omnivores dropped abruptly in 2014

Explain regime shifts



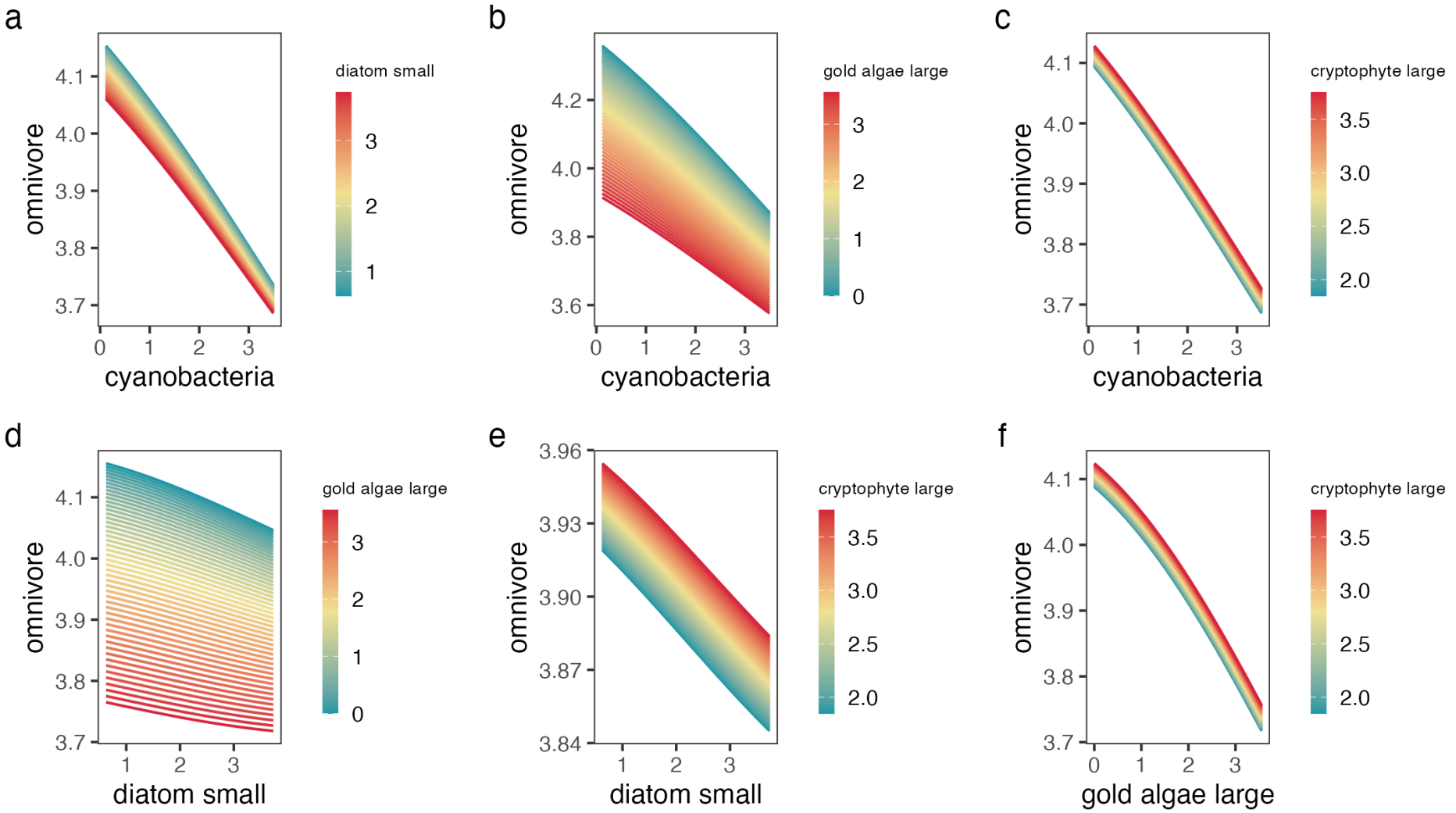


LOO R^2 = **0.81**

Omnivores ↓ with cyanobacteria, small diatoms, large gold algae

Prey hypothesis: **not supported**

Explain regime shifts: partial dependence





x: predictor 1
lines: predictor 2

Cyanobacteria ↓ omnivores

Lines never cross

Propagation of regime shifts

1 · Real predictor



2 candidate pairs

0 testable

Predictor shifted **after**

2 · Shift-free surrogate (× 100)



Gap between real and surrogate prediction = evidence for propagation

No evidence yet: too few shifts

Conclusions



Shifts are detectable



Biotic drivers matter



Propagation, prediction: open

Take home message

Plankton regime shifts can be detected and partly explained by biotic drivers. Whether they propagate and can be predicted is still open.



Acknowledgments

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Kofinanziert von der Europäischen Union

igkb
Internationale Gewässerschutzkommission für den Bodensee

IBKF

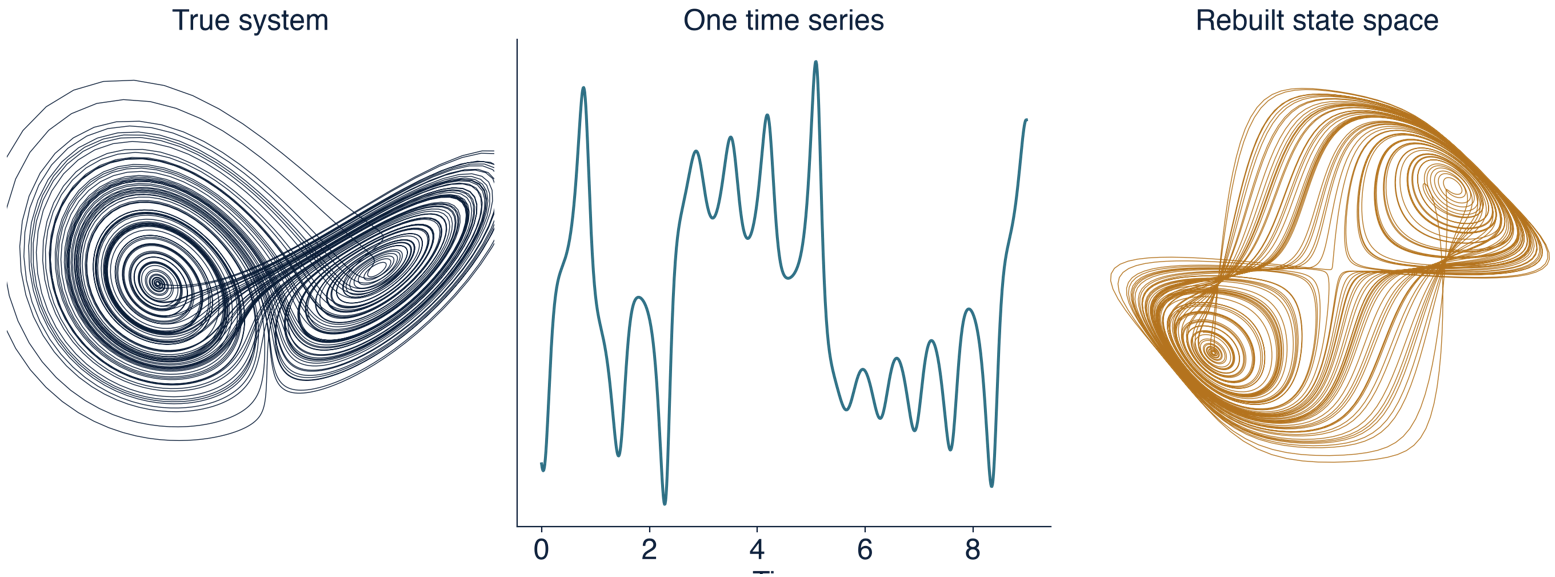
SeeWandel-Climate · Pomati lab and colleagues at Eawag
Luis J. Gilarranz · Stefanie Eyring · Stephan B. Munch · Tanya L. Rogers · Francesco Pomati
Data: IGKB, IBKF, University of Konstanz (D. Straile)

Supplementary material

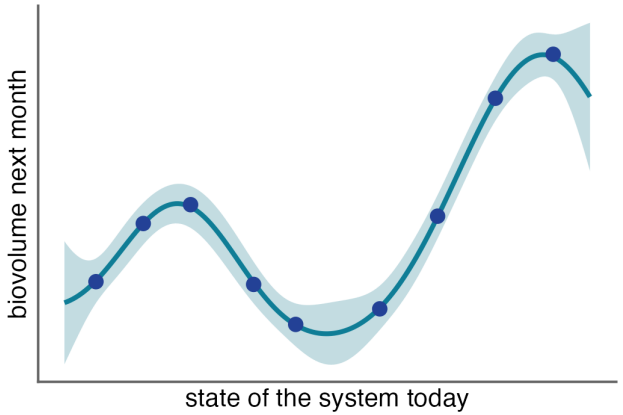
Methods in detail



What is GP-EDM?



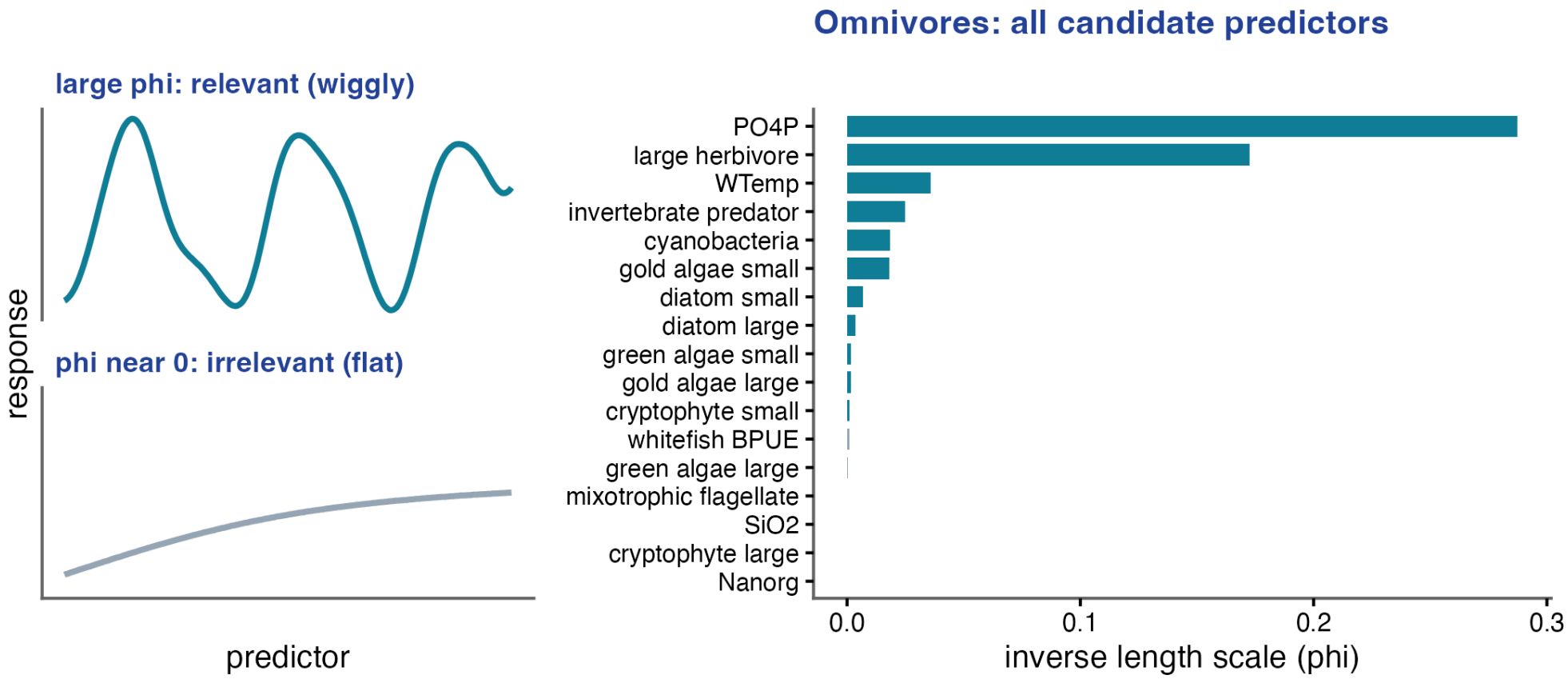
Gaussian process: no fixed equation



Next month = f(past states, drivers)

f learned from the data: Gaussian process

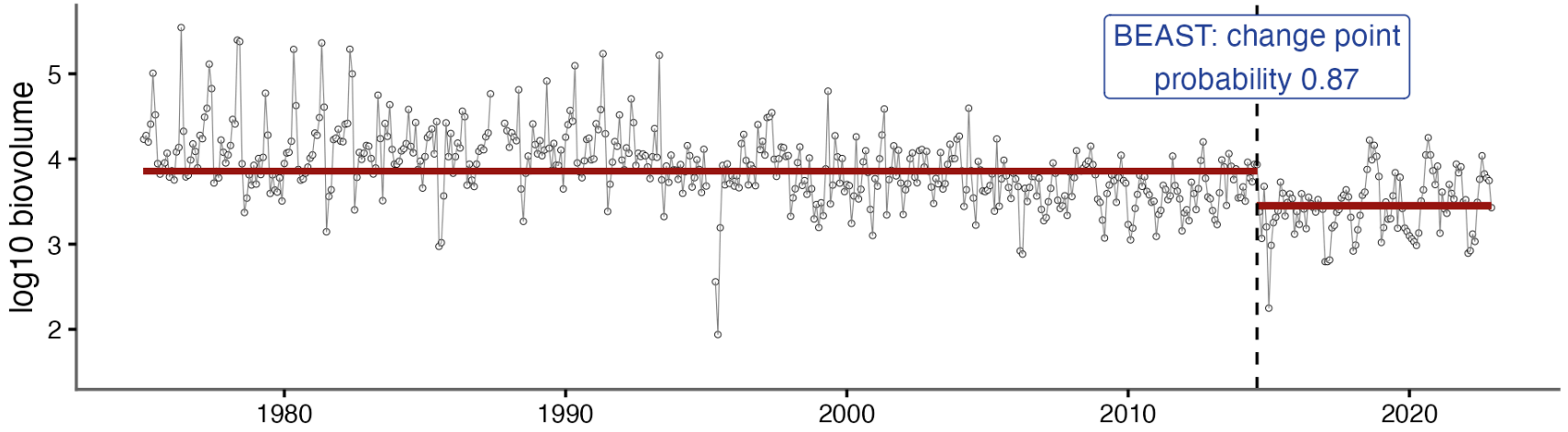
Automatic relevance determination (ARD)



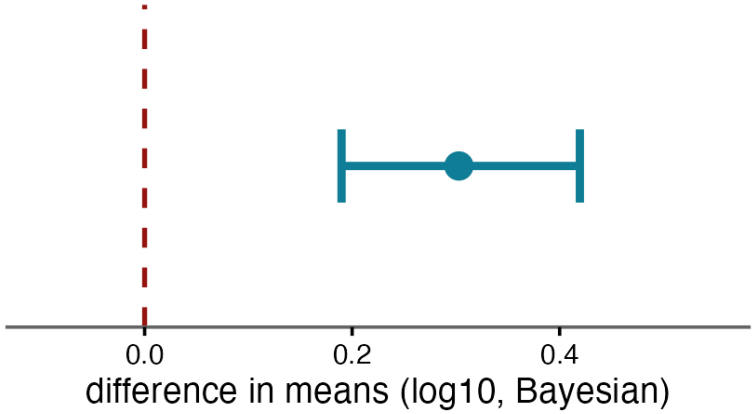
One length scale per predictor: **phi near 0** = predictor dropped

Detecting regime shifts: BEAST and two criteria

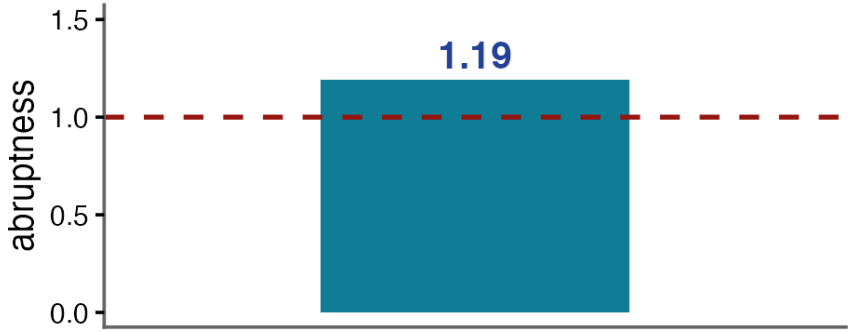
1: Candidate: BEAST change point



2: 95 % interval excludes 0



3: Abruptness > 1



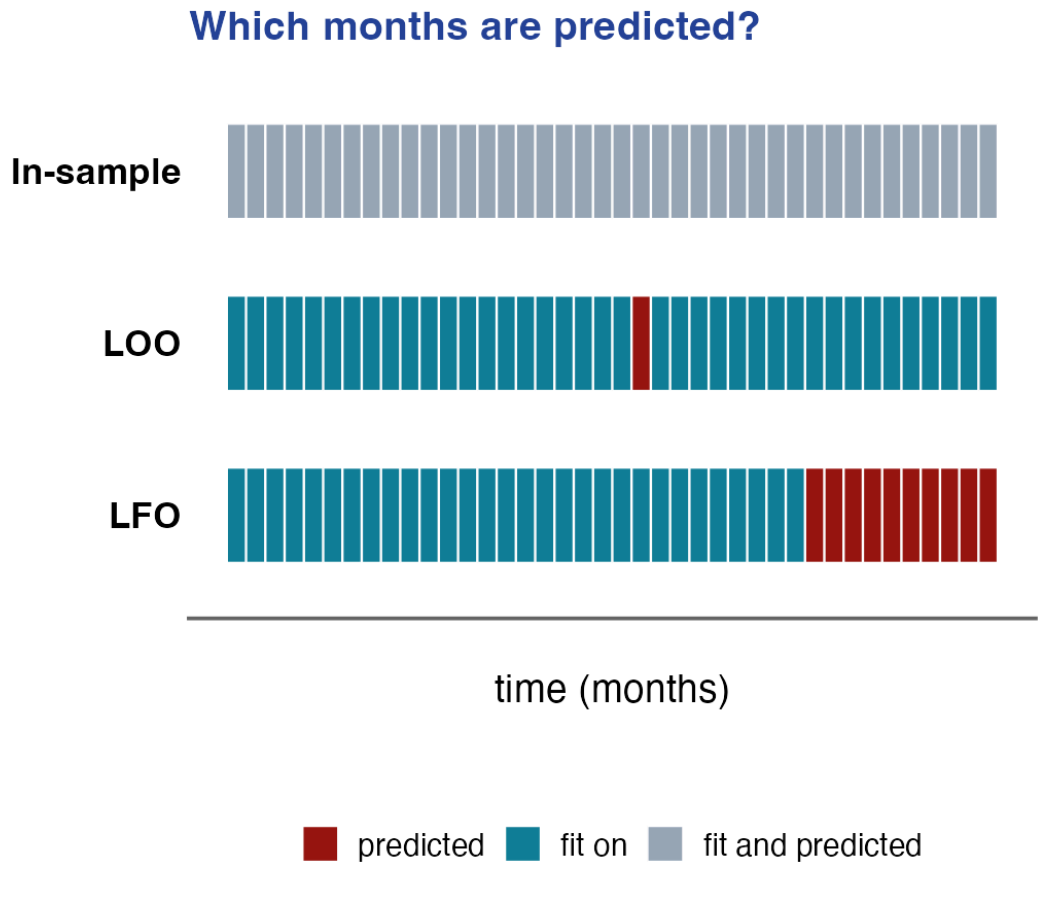
Kept if **both** hold

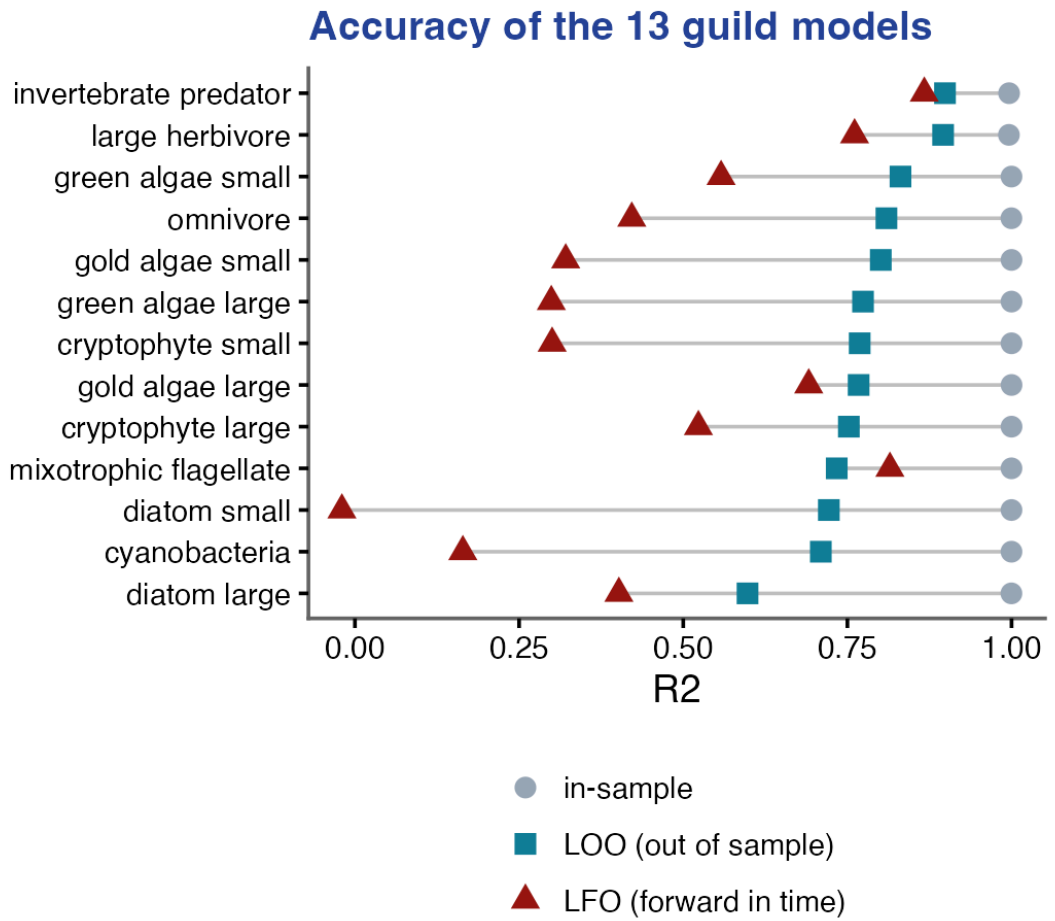
2 interval excludes 0

3 abruptness > 1

abruptness = |difference| / mean SD of the two periods

In-sample versus out-of-sample R^2





In-sample: $R^2 = 1$, a GP interpolates

LOO: one month left out

LFO: forecast the last 10 %

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Figure of Lake Constance (map, phosphorus, temperature): D. Straile, University of Konstanz. Data: IGKB, IBKF.